

# Week 4 Assignment

Ellen Bledsoe

2026-09-15

## 2. Portal Data Aggregation (15 pts)

[1] "2a"

```
# A tibble: 49 x 2
  species_id count
  <chr>      <int>
1 AB         303
2 AH         437
3 AS           2
4 BA          46
5 CB          50
6 CM          13
7 CQ          16
8 CS           1
9 CT           1
10 CU           1
# i 39 more rows
```

[1] "2b"

```
# A tibble: 535 x 3
# Groups:   species_id [49]
  species_id year count
  <chr>      <dbl> <int>
1 AB        1980     5
2 AB        1981     7
3 AB        1982    34
4 AB        1983    41
5 AB        1984    12
```

```

6 AB      1985    14
7 AB      1986     5
8 AB      1987    35
9 AB      1988    39
10 AB     1989    31
# i 525 more rows

```

```
[1] "2c"
```

```

# A tibble: 26 x 2
  year avg_mass
  <dbl>   <dbl>
1  1977   42.7
2  1978   45
3  1979   45.9
4  1980   48.1
5  1981   49.1
6  1982   47.9
7  1983   47.2
8  1984   48.4
9  1985   48.0
10 1986   49.4
# i 16 more rows

```

### 3. Shrub Volume Aggregation (15 pts)

```
[1] "3a"
```

```

# A tibble: 3 x 2
  experiment avg_height
  <dbl>       <dbl>
1         1         4.7
2         2        5.12
3         3        3.85

```

```
[1] "3b"
```

```

# A tibble: 3 x 3
  experiment avg_height max_height
  <dbl>       <dbl>       <dbl>
1         1         4.7         4.7
2         2        5.12        5.12
3         3        3.85        3.85

```

|   |   |      |     |
|---|---|------|-----|
| 1 | 1 | 4.7  | 9.6 |
| 2 | 2 | 5.12 | 7.6 |
| 3 | 3 | 3.85 | 7.5 |

[1] "3c"

```
# A tibble: 5 x 4
  site avg_height max_height min_height
<dbl> <dbl> <dbl> <dbl>
1     1     6.47     9.6     2.2
2     2     2.83     4     1.5
3     3     4.77     7.5     2.3
4     4     4.13     6.5     2.7
5     5     5.2     5.2     5.2
```

#### 4. Portal Data Joins (25 pts)

[1] "4a"

```
# A tibble: 34,786 x 12
  record_id month   day  year plot_id species_id sex hindfoot_length weight
  <dbl> <dbl> <dbl> <dbl> <dbl> <chr> <chr> <dbl> <dbl>
1         1     7    16  1977     2 NL      M      32    NA
2         2     7    16  1977     3 NL      M      33    NA
3         3     7    16  1977     2 DM      F      37    NA
4         4     7    16  1977     7 DM      M      36    NA
5         5     7    16  1977     3 DM      M      35    NA
6         6     7    16  1977     1 PF      M      14    NA
7         7     7    16  1977     2 PE      F      NA    NA
8         8     7    16  1977     1 DM      M      37    NA
9         9     7    16  1977     1 DM      F      34    NA
10        10     7    16  1977     6 PF      F      20    NA
```

# i 34,776 more rows

# i 3 more variables: genus <chr>, species <chr>, taxa <chr>

[1] "4b"

```
# A tibble: 34,786 x 13
  record_id month   day  year plot_id species_id sex hindfoot_length weight
  <dbl> <dbl> <dbl> <dbl> <dbl> <chr> <chr> <dbl> <dbl>
```

|    |    |   |    |      |   |    |   |    |    |
|----|----|---|----|------|---|----|---|----|----|
| 1  | 1  | 7 | 16 | 1977 | 2 | NL | M | 32 | NA |
| 2  | 2  | 7 | 16 | 1977 | 3 | NL | M | 33 | NA |
| 3  | 3  | 7 | 16 | 1977 | 2 | DM | F | 37 | NA |
| 4  | 4  | 7 | 16 | 1977 | 7 | DM | M | 36 | NA |
| 5  | 5  | 7 | 16 | 1977 | 3 | DM | M | 35 | NA |
| 6  | 6  | 7 | 16 | 1977 | 1 | PF | M | 14 | NA |
| 7  | 7  | 7 | 16 | 1977 | 2 | PE | F | NA | NA |
| 8  | 8  | 7 | 16 | 1977 | 1 | DM | M | 37 | NA |
| 9  | 9  | 7 | 16 | 1977 | 1 | DM | F | 34 | NA |
| 10 | 10 | 7 | 16 | 1977 | 6 | PF | F | 20 | NA |

# i 34,776 more rows

# i 4 more variables: genus <chr>, species <chr>, taxa <chr>, plot\_type <chr>

[1] "4c"

# A tibble: 15,660 x 10

|    | record_id | month | day   | year  | plot_id | species_id | sex   | hindfoot_length | weight |
|----|-----------|-------|-------|-------|---------|------------|-------|-----------------|--------|
|    | <dbl>     | <dbl> | <dbl> | <dbl> | <dbl>   | <chr>      | <chr> | <dbl>           | <dbl>  |
| 1  | 1         | 7     | 16    | 1977  | 2       | NL         | M     | 32              | NA     |
| 2  | 3         | 7     | 16    | 1977  | 2       | DM         | F     | 37              | NA     |
| 3  | 7         | 7     | 16    | 1977  | 2       | PE         | F     | NA              | NA     |
| 4  | 14        | 7     | 16    | 1977  | 8       | DM         | <NA>  | NA              | NA     |
| 5  | 16        | 7     | 16    | 1977  | 4       | DM         | F     | 36              | NA     |
| 6  | 18        | 7     | 16    | 1977  | 2       | PP         | M     | 22              | NA     |
| 7  | 19        | 7     | 16    | 1977  | 4       | PF         | <NA>  | NA              | NA     |
| 8  | 20        | 7     | 17    | 1977  | 11      | DS         | F     | 48              | NA     |
| 9  | 21        | 7     | 17    | 1977  | 14      | DM         | F     | 34              | NA     |
| 10 | 28        | 7     | 17    | 1977  | 11      | DM         | M     | 38              | NA     |

# i 15,650 more rows

# i 1 more variable: plot\_type <chr>

## 5. Portal Data dplyr Review (25 pts)

[1] "5a"

# A tibble: 19,344 x 5

|   | year  | genus     | species  | weight | plot_type                |
|---|-------|-----------|----------|--------|--------------------------|
|   | <dbl> | <chr>     | <chr>    | <dbl>  | <chr>                    |
| 1 | 1977  | Dipodomys | merriami | 40     | Long-term Krat Exclosure |
| 2 | 1977  | Dipodomys | merriami | 29     | Control                  |
| 3 | 1977  | Dipodomys | merriami | 46     | Control                  |

```

4 1977 Dipodomys ordii 52 Control
5 1977 Perognathus flavus 8 Control
6 1977 Onychomys sp. 22 Long-term Krat Exclosure
7 1977 Perognathus flavus 7 Control
8 1977 Dipodomys merriami 22 Control
9 1977 Perognathus flavus 8 Control
10 1977 Dipodomys merriami 41 Control
# i 19,334 more rows

```

```
[1] "5b"
```

```
# A tibble: 52 x 5
```

```
# Groups:   year [26]
```

|    | year  | plot_type                | min_weight | max_weight | mean_weight |
|----|-------|--------------------------|------------|------------|-------------|
|    | <dbl> | <chr>                    | <dbl>      | <dbl>      | <dbl>       |
| 1  | 1977  | Control                  | 6          | 149        | 50.4        |
| 2  | 1977  | Long-term Krat Exclosure | 7          | 50         | 34.8        |
| 3  | 1978  | Control                  | 6          | 223        | 70.8        |
| 4  | 1978  | Long-term Krat Exclosure | 6          | 232        | 35.9        |
| 5  | 1979  | Control                  | 7          | 274        | 68.1        |
| 6  | 1979  | Long-term Krat Exclosure | 6          | 122        | 24.4        |
| 7  | 1980  | Control                  | 5          | 214        | 66.2        |
| 8  | 1980  | Long-term Krat Exclosure | 5          | 155        | 26.5        |
| 9  | 1981  | Control                  | 4          | 264        | 68.0        |
| 10 | 1981  | Long-term Krat Exclosure | 4          | 195        | 34.7        |

```

# i 42 more rows

```

## 6. Shrub Volume Bind (10 pts)

```
# A tibble: 15 x 7
```

|   | site  | experiment | length | width | height | respiratory_rate | average_temp_C |
|---|-------|------------|--------|-------|--------|------------------|----------------|
|   | <dbl> | <dbl>      | <dbl>  | <dbl> | <dbl>  | <dbl>            | <dbl>          |
| 1 | 1     | 1          | 2.2    | 1.3   | 9.6    | 2.2              | 15.1           |
| 2 | 1     | 2          | 2.1    | 2.2   | 7.6    | 4                | 20.2           |
| 3 | 1     | 3          | 2.7    | 1.5   | 2.2    | 6.1              | 24.7           |
| 4 | 2     | 1          | 3      | 4.5   | 1.5    | 2.3              | 15.2           |
| 5 | 2     | 2          | 3.1    | 3.1   | 4      | 4.1              | 22             |
| 6 | 2     | 3          | 2.5    | 2.8   | 3      | 6.2              | 25.1           |
| 7 | 3     | 1          | 1.9    | 1.8   | 4.5    | 1.8              | 14.2           |
| 8 | 3     | 2          | 1.1    | 0.5   | 2.3    | 3.5              | 19             |
| 9 | 3     | 3          | 3.5    | 2     | 7.5    | 5.7              | 23.6           |

|    |   |   |     |     |     |     |      |
|----|---|---|-----|-----|-----|-----|------|
| 10 | 4 | 1 | 2.9 | 2.7 | 3.2 | 1.9 | 14.9 |
| 11 | 4 | 2 | 4.5 | 4.8 | 6.5 | 3.5 | 20.3 |
| 12 | 4 | 3 | 1.2 | 1.8 | 2.7 | 5.8 | 24.1 |
| 13 | 5 | 1 | 2.6 | 0.8 | NA  | 2   | 19.2 |
| 14 | 5 | 2 | 1.8 | NA  | 5.2 | 4.7 | 22.7 |
| 15 | 5 | 3 | 3.1 | 2.2 | NA  | 6.2 | 25   |

## 7. Shrub Volume Join (15 pts)

[1] "7a"

# A tibble: 15 x 6

|    | site  | experiment | length | width | height | manipulation |
|----|-------|------------|--------|-------|--------|--------------|
|    | <dbl> | <dbl>      | <dbl>  | <dbl> | <dbl>  | <chr>        |
| 1  | 1     | 1          | 2.2    | 1.3   | 9.6    | control      |
| 2  | 1     | 2          | 2.1    | 2.2   | 7.6    | burn         |
| 3  | 1     | 3          | 2.7    | 1.5   | 2.2    | rainout      |
| 4  | 2     | 1          | 3      | 4.5   | 1.5    | control      |
| 5  | 2     | 2          | 3.1    | 3.1   | 4      | burn         |
| 6  | 2     | 3          | 2.5    | 2.8   | 3      | rainout      |
| 7  | 3     | 1          | 1.9    | 1.8   | 4.5    | control      |
| 8  | 3     | 2          | 1.1    | 0.5   | 2.3    | burn         |
| 9  | 3     | 3          | 3.5    | 2     | 7.5    | rainout      |
| 10 | 4     | 1          | 2.9    | 2.7   | 3.2    | control      |
| 11 | 4     | 2          | 4.5    | 4.8   | 6.5    | burn         |
| 12 | 4     | 3          | 1.2    | 1.8   | 2.7    | rainout      |
| 13 | 5     | 1          | 2.6    | 0.8   | NA     | control      |
| 14 | 5     | 2          | 1.8    | NA    | 5.2    | burn         |
| 15 | 5     | 3          | 3.1    | 2.2   | NA     | rainout      |

[1] "7b"

# A tibble: 12 x 9

|   | site  | experiment | length | width | height | manipulation | latitude | longitude |
|---|-------|------------|--------|-------|--------|--------------|----------|-----------|
|   | <dbl> | <dbl>      | <dbl>  | <dbl> | <dbl>  | <chr>        | <dbl>    | <dbl>     |
| 1 | 1     | 1          | 2.2    | 1.3   | 9.6    | control      | 29.6     | -82.3     |
| 2 | 1     | 2          | 2.1    | 2.2   | 7.6    | burn         | 29.6     | -82.3     |
| 3 | 1     | 3          | 2.7    | 1.5   | 2.2    | rainout      | 29.6     | -82.3     |
| 4 | 2     | 1          | 3      | 4.5   | 1.5    | control      | 29.3     | -82.4     |
| 5 | 2     | 2          | 3.1    | 3.1   | 4      | burn         | 29.3     | -82.4     |
| 6 | 2     | 3          | 2.5    | 2.8   | 3      | rainout      | 29.3     | -82.4     |

|    |   |   |     |     |     |         |      |       |
|----|---|---|-----|-----|-----|---------|------|-------|
| 7  | 3 | 1 | 1.9 | 1.8 | 4.5 | control | 29.8 | -82.2 |
| 8  | 3 | 2 | 1.1 | 0.5 | 2.3 | burn    | 29.8 | -82.2 |
| 9  | 3 | 3 | 3.5 | 2   | 7.5 | rainout | 29.8 | -82.2 |
| 10 | 4 | 1 | 2.9 | 2.7 | 3.2 | control | 30.0 | -82.6 |
| 11 | 4 | 2 | 4.5 | 4.8 | 6.5 | burn    | 30.0 | -82.6 |
| 12 | 4 | 3 | 1.2 | 1.8 | 2.7 | rainout | 30.0 | -82.6 |

# i 1 more variable: elevation <dbl>